

Problem Statement & Scope of Work

1. Project Title

****Maths Quiz Application****

2. Problem Statement

Many students and beginner programmers looking to test basic arithmetic abilities lack simple, focused, and distraction-free command-line tools. Many existing educational applications are either bloated with unnecessary interfaces, require heavy external installations, or run inside complex web environments.

There is a clear requirement for a lightweight, console-driven mathematics evaluation tool that can:

- Run natively on any machine with standard Python 3.
- Provide a clean, distraction-free environment for foundational calculation practice.
- Validate multiple-choice arithmetic answers instantly.
- Accurately compute and display the user's score upon quiz termination.

3. Project Objectives

- ****Interactive Console Flow:**** Collect user credentials (name) and display a clear topic menu.
- ****Multi-domain Coverage:**** Offer arithmetic problems across four fundamental operations:
 1. Addition
 2. Subtraction
 3. Multiplication
 4. Division
- ****Robust Answer Checking:**** Compare user responses against preset keys and distinguish between incorrect answers and invalid option entries.
- ****Performance Evaluation:**** Dynamically maintain and report an overall score counter.

4. Scope of the Project

- ****Current Scope:****
 - Implements procedural programming using Python control structures ('if', 'elif', 'else').
 - Terminal-based execution without reliance on third-party libraries.
 - Interactive multiple-choice format (MCQ) covering standard operations.
- ****Out of Scope (Current Release):****
 - Graphical user interface (GUI).
 - Web-based deployment or database-backed authentication.

5. Expected Outcomes

- An executable, error-free Python script ('quiz.py') running in any terminal shell.
- Clear and immediate feedback for each user attempt.
- Accurate real-time scoring summarizing the user's arithmetic proficiency.



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